

Predicting League of Legends Match Outcomes from Early-Game Signals

Liam Sandy*

Department of Computational Mathematics, Science, and Engineering

Michigan State University, East Lansing, MI 48824

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Abstract

League of Legends is a multiplayer online battle arena (MOBA) game where competitive advantage can shift dramatically within minutes. Analysts and coaching staffs often rely on intuition rather than data-driven insight to interpret the early phases of play. This project proposes a machine learning pipeline to predict match outcomes using only the first ten minutes of in game telemetry from Diamond ranked matches. The workflow will explore three supervised models—penalized logistic regression, random forest, and a small multilayer perceptron—evaluating which best captures the relationship between early gold, experience, and objective control and the eventual winner. The outcome will be a reproducible modeling framework, accuracy and interpretability comparison, and a concise presentation highlighting how quantitative models can support strategic analysis in esports.

GitHub Repository: https://github.com/liamsandy/cmse492_project

BACKGROUND AND MOTIVATION

League of Legends (LoL) hosts one of the world’s largest competitive esports ecosystems, with millions of players and organized tournaments worldwide. Each match generates extensive telemetry on kills, gold, objectives, and map control—data that analysts must interpret quickly to guide coaching decisions or live commentary. Early game prediction is particularly valuable, as production crews and analysts benefit from tools that estimate which team is more likely to win before mid game objectives unfold.

Traditional approaches to match prediction rely on full game summaries or draft phase information. However, focusing exclusively on early game metrics allows for real time assessment. Machine learning is well suited for this because it can model nonlinear dependencies across multiple performance indicators simultaneously. This project aims to quantify how much predictive signal is contained in early metrics alone and to create a transparent model comparison pipeline that demonstrates interpretable, reproducible esports analytics.

DATA DESCRIPTION

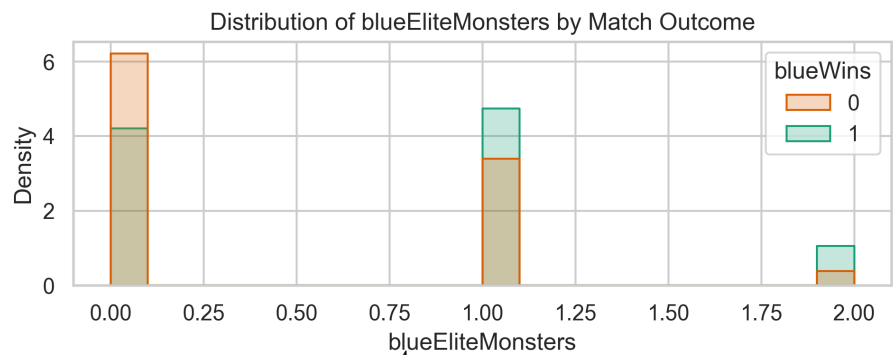
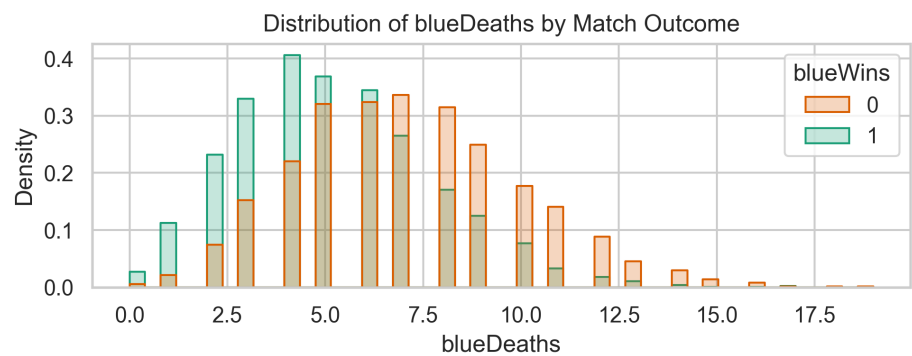
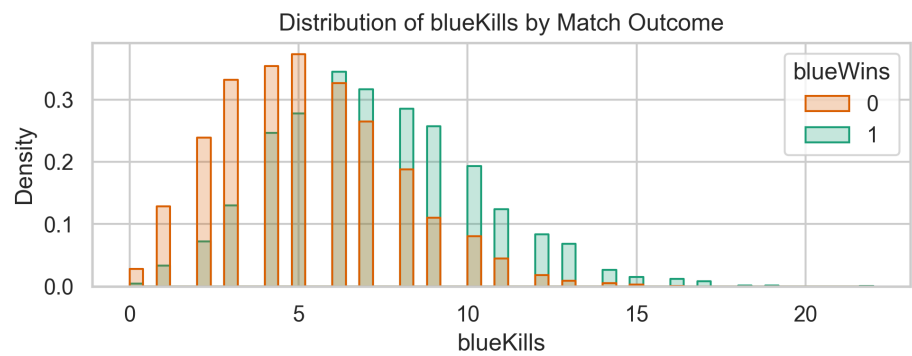
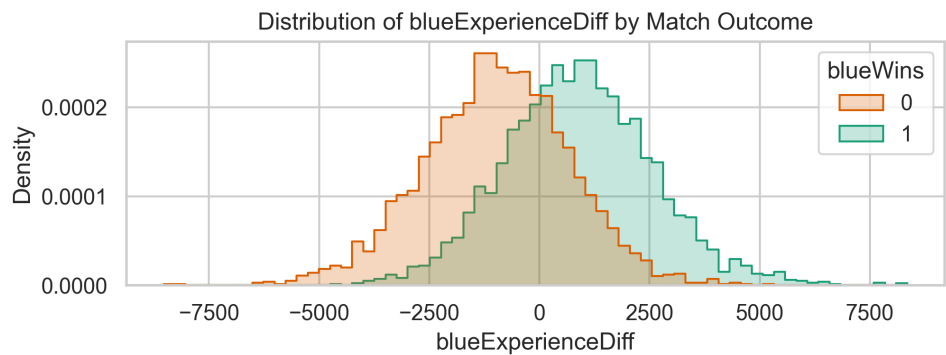
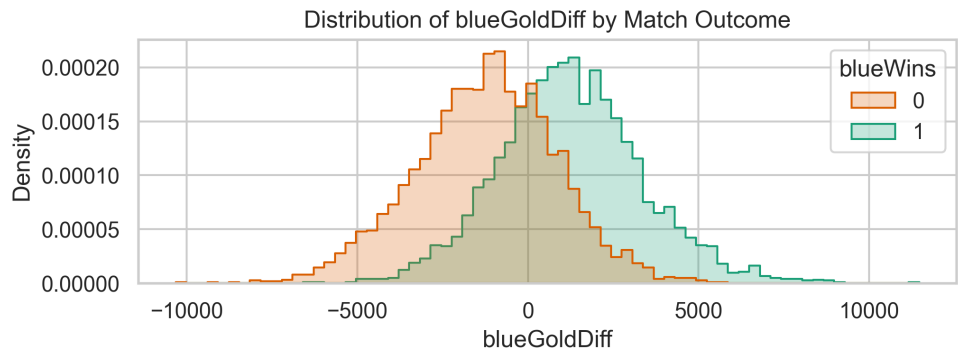
Data Source

The data used in this project originates from the *League of Legends Diamond Ranked Games (10 min)* dataset publicly available on Kaggle. It aggregates 9,879 matches from the 2020 ranked season at the Diamond skill tier, representing roughly the top 2.5% of players. Each record captures team level performance at the ten minute mark, including gold, experience, kills, deaths, wards, and early neutral objectives such as dragons and heralds. The dataset was compiled using Riot Games' official match API, ensuring reliable and verified telemetry.

Structure and Quality

- **Rows:** 9,879 matches
- **Columns:** 39 features plus a binary target (`blueWins`)
- **Target:** 1 if the Blue team wins, 0 otherwise

No missing values or nulls are present. Classes are balanced (approximately 50% Blue wins). Outlier checks using interquartile range thresholds show less than 2% of data outside normal bounds, meaning minimal preprocessing is required beyond scaling and optional log transformation for skewed numeric fields.



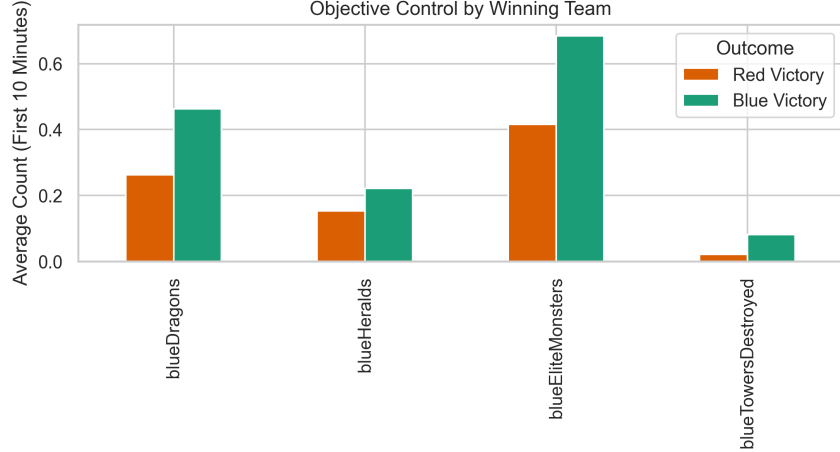


FIG. 2: Average early game objective control (dragons, heralds, turrets) by team outcome. Blue side wins show higher early map control.

Planned Preprocessing

- **Split:** 80/20 stratified train test split.
- **Scaling:** StandardScaler for linear and neural models.
- **Transformations:** Log scale gold and experience variables to reduce skew.
- **Feature Selection:** Drop identifiers and low variance metrics; consider derived ratios such as gold per minute.

PROPOSED METHODOLOGY

Overview

The problem is formulated as a supervised binary classification task:

$$f(\mathbf{x}) = \hat{y}, \quad \hat{y} = \begin{cases} 1 & \text{Blue team wins} \\ 0 & \text{otherwise.} \end{cases}$$

Three model tiers will be benchmarked:

1. **Penalized Logistic Regression** – establishes a baseline with interpretable coefficients.

2. **Random Forest Classifier** – captures nonlinear thresholds and feature interactions.
3. **Multilayer Perceptron (MLP)** – models more complex relationships through dense layers with dropout.

Justification

Each model provides a balance between interpretability and predictive power:

- Logistic regression provides explainable coefficients.
- Random forest highlights key non linear patterns and feature importances.
- MLP extends performance potential while maintaining compact architecture.

Hyperparameters for each model will be tuned via five fold cross validation. Metrics and configuration results will be stored in reproducible JSON logs.

EVALUATION FRAMEWORK

Metrics

Three metrics will be emphasized:

- **Accuracy** – share of correct predictions.
- **F1 Score** – harmonic mean of precision and recall for balanced evaluation.
- **ROC-AUC** – measures rank order quality across thresholds.

ROC-AUC serves as the main metric since it reflects ranking ability rather than a fixed threshold.

Validation Strategy

- **Split:** 80/20 stratified split.
- **Cross-Validation:** Five-fold CV within the training set.

- **Baseline:** Majority-class predictor for context.

A successful model should surpass 0.80 ROC-AUC and maintain consistent results across folds.

TIMELINE AND MILESTONES

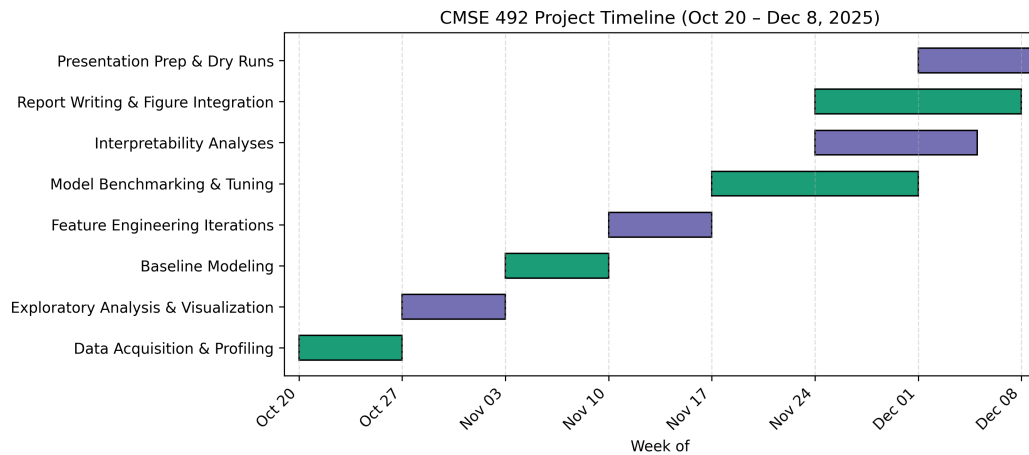


FIG. 3: Planned Gantt chart for weekly progress and milestones through December 2025.

Narrative Summary: Weeks 1–2: exploratory data analysis and preprocessing pipeline setup. Weeks 3–5: model development, tuning, and comparison. Weeks 6–7: interpretability analysis (feature importance and SHAP values). Weeks 8–9: final report and presentation preparation. A buffer week before the December 8 submission will handle last minute debugging and figure revisions.

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* sandylia@msu.edu